
UNIFIED MECHANICS

Volume V

Formal Register, Methods, and Reproducibility

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VOLUME V OF V

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A CORRIGIBLE EDITION. This text is corrigible. Earlier manifests and printings are retained as provenance, not as immutable authority. Corrections are recorded by version and supersession history; no scientific claim is frozen.

ABSTRACT

This volume is the audit layer: axioms and definitions, dependency graph, scientifically regraded results, computational certificates, empirical protocols, source crosswalk, supersession history, and open-obligation register. It makes the series inspectable without imposing an immutable scientific state.

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The living scientific record

1.1 Purpose of the register

This volume is the audit surface for the series. It does not add another layer of physical ontology. It records the commitments on which the argument depends, the meanings of its terms, the scientific grade of each inherited theorem, the relation between source documents and rewritten chapters, and the conditions under which a result must be revised or abandoned.

A register is necessary because the corpus developed through books, technical papers, addenda, computational notebooks, generated tables, data files, manifests, and several editorial states. Later documents often corrected earlier ones, while exact duplicates appeared in different formats or directories. Preserving everything therefore cannot mean repeating every sentence. It means preserving each distinct claim, equation, dataset, counterexample, certificate, failure, and historical transition in a structure where its current disposition is visible.

1.2 No-freeze versioning

Scientific content is never frozen. At most, an analysis protocol may be pre-registered for the duration of a test. The living edition separates four operations:

Version

a citable state of the manuscript, code, data, and register;

Correction

a change that repairs an error without changing the intended claim;

Supersession

a new claim or derivation that replaces an earlier one while retaining its provenance;

Withdrawal

a route that no longer supports the stated conclusion, retained so that it cannot silently return.

No manifest may declare a proposition scientifically immutable. Reproduc-

tion means that a result can be regenerated from a declared historical state; it does not require future editions to continue believing the result.

EDITORIAL RULE 1.1 — PROVENANCE WITHOUT AUTHORITY

Every superseded or withdrawn item remains addressable by source path and digest. Its survival in the record establishes what was written and tested, not that the claim is true. The current scientific grade is governed by the latest explicit regrading, which may itself later be superseded.

1.3 Corpus census

The combined audit covered the repository and the larger SCI archive. It indexed 1,319 files, representing 791 distinct SHA-256 hashes. Of these, 491 unique text-bearing sources were extracted into the claim-and-coverage ledger; 155 were classified as substantive scientific sources and 132 as manuscript files. Exact duplicate groups and cross-format renderings were consolidated, while their paths were retained. No text extraction failed in the final index.

The 491-source crosswalk contains 148 implementation files, 126 manuscripts, 105 data or certificate files, 64 supporting texts, 23 orientation or ledger files, 19 manifests, and 6 historical manuscripts. The machine-readable ledger records every source's full digest, format, role, approximate word count, leading topics, claim labels, assigned volume and section, and review status.

Scientific grades and inference rules

2.1 A typed claim system

The inherited corpus frequently used `PROVED` for several logically different achievements. This edition replaces that single word with a typed system.

TABLE 2.1. Scientific grades used throughout the living edition.

Grade	Criterion
Definition	Fixes vocabulary or a mathematical object. It is neither true nor false until judged for consistency and usefulness.
Framework postulate	A declared starting commitment. Consequences remain conditional on it.
Exact identity	Follows by algebra from displayed definitions, with no physical identification implied.
Internal lemma/theorem	Follows from stated formal hypotheses by a checkable proof.
Computational result	Regenerated by a declared algorithm and certificate; scope is limited by finite precision, enumeration completeness, and implementation assumptions.
Established external result	A published theorem or observation imported with its actual hypotheses, model, units, and uncertainty.
Conditional identification	Connects an internal object with a physical one only after a named realisation map is granted.
Interpretation	A coherent reading that does not itself add a testable dynamical consequence.
Conjecture	A mathematical or physical statement for which the required proof or mechanism is absent.
Forecast	A predeclared output exposed to future or withheld data.
Open obligation	A missing bridge whose completion is required by a stronger claim.

Grade	Criterion
Withdrawn	A retained historical route that no longer supports its conclusion.

2.2 The claim tuple

Each serious claim should be representable as

$$\mathfrak{C} = (S, H, D, M, R, G, F), \quad (2.1)$$

where S is the statement, H its hypotheses, D the derivation or data, M the measurement map, R the reproducible artefact, G the grade, and F a falsifier or supersession condition. A purely mathematical identity may leave M empty. A physical claim may not.

The following invalid moves recur in speculative theory and are explicitly prohibited:

1. promoting an exact identity to a physical law without a realisation map;
2. treating numerical agreement after model selection as a forecast;
3. using a standard result as evidence for the framework merely because the framework can restate it;
4. counting correlated parameters as independent confirmations;
5. declaring a missing mechanism “forced” by a verbal definition;
6. changing conventions, datasets, stage counts, or branches after viewing the residual without recording the selection;
7. interpreting failure to falsify as confirmation.

2.3 Dependency and proof discipline

For each theorem, the proof graph must distinguish four edge types:

$$\begin{aligned} &\text{definition} \longrightarrow \text{formal consequence} \\ &\longrightarrow \text{realisation assumption} \longrightarrow \text{empirical readout.} \end{aligned} \quad (2.2)$$

An external theorem enters only at the edge licensed by its hypotheses. For example, lattice classification can identify E_8 after rank, integrality, evenness, unimodularity, and positive definiteness have been supplied; it cannot supply those premises. Gleason’s theorem constrains probability measures under non-contextual additivity hypotheses (Gleason, 1957; Busch, 2003; Caves et al., 2004); it does not derive those hypotheses from fairness language. Wigner and Stone

constrain transformations only after transition-probability preservation, bijectivity, and continuity have been stated (Wigner, 1959; Stone, 1932).

Commitments and definitions

3.1 Commitments A1–A6

The six inherited axioms are preserved below with their living-edition status.

TABLE 3.1. Foundational commitments.

ID	Name and grade	Scientific statement
A1	Systemhood; framework postulate	A model may treat a system as one entity while permitting internally distinguishable states or parts. This is a modelling convention, not an empirical law.
A2	Ground; ontological postulate	The first distinction is said to constitute its own “where”. No spacetime consequence follows until a mathematical state space and geometry are constructed.
A3	Two-sided reading; framework postulate	A mixed relation is counted in both orientations. This supplies the factor of two in the retained-fraction normalization; all downstream numerical claims remain conditional on it. It does not by itself prove evenness or lattice self-duality.
A4	Saturation; open physical postulate	A single accumulating channel is assumed to reach finite capacity. Capacity, threshold, state space, and dynamics remain unspecified.
A5	Conservation of readout; conservation postulate	Continuing state, outgoing record, and their order are assumed to conserve a declared total. Physical use requires an explicit conserved measure or current.

ID	Name and grade	Scientific statement
A6	Admissible reduction; definition	A reduction is declared unital, idempotent, completely positive, bimodular over the readable algebra, and covariant under equivalent carrier descriptions. Operator-algebra consequences require a specified algebra, action, norm, and fixed space.

3.2 Definitions D1–D18

TABLE 3.2. Framework vocabulary, preserved and delimited.

ID	Term	Living-edition definition
D1	Distinction	A relation or property that prevents complete equivalence under the comparison being made.
D2	Identity	Recoverability of declared distinctions through an admissible family of changes.
D3	Relation	A rule by which distinct contents compare, constrain, or transform one another without automatic identification.
D4	Seven doings	Distinguish, hold, close, answer, change, order, and record: the corpus’s constitutive grammar.
D5	Traversal stages	Persist, distinguish, relate, propagate, resolve, return, and close: a process-oriented rendering of D4.
D6	Object	A stable equivalence class of closed traversals relative to a declared observation and timescale.
D7	Update	The two-channel map $(a, b) \mapsto (a + b, a)$.
D8	Channels	The outward share u and retained share r , normalized by $u + r = 1$.
D9	Three weights	$W_L = u^2$, $W_B = 2ur$, and $W_M = r^2$. The physical names are interpretations.
D10	Crossing	One enacted step of self-description across a declared channel. Its assignment of cost r is a separate conjectural rule.

ID	Term	Living-edition definition
D11	Record	A distinction that remains discriminable and recoverable under the future dynamics relevant to the task. The inherited binary criterion is an idealization; physical records can be noisy or graded.
D12	Observer	A system in which another system's distinction becomes coupled and persistently recorded; a mechanical observer includes preparation, coupling, propagation, resolution, amplification, and record retention.
D13	Carrier	The structure on which the framework writes self-describing records. Self-duality is a modelling requirement, not a derivation of an even unimodular lattice.
D14	Primitive cross- ing	A smallest nonzero displacement in a chosen carrier model; in the E_8 realisation these are roots of squared norm two.
D15	Cluster projec- tion	A map C from carrier observables to a readable subalgebra satisfying A6.
D16	Decay generator	An operator K whose kernel is the readable algebra; the minimal choice is $I - C$, while a general positive generator may have a richer spectrum.
D17	Readable alge- bra	The subalgebra fixed by the declared reduction and therefore retained by the effective description. Its identification with physical observables is model-dependent.
D18	Geometry projectors	Orthogonal projectors P_X, P_T, P_I assigned to external distinction, causal order, and closure, with $P_X + P_T + P_I$ the geometry block. Their ranks are assumptions until represented independently.

Regraded formal register T1–T39

The following table is the controlling scientific register. It preserves the inherited numbering so every earlier proof, certificate, and cross-reference remains traceable. The short ruling is not a replacement for the fuller derivations in Volumes I–IV; it states the maximum defensible grade.

TABLE 4.1. Living-edition regrade of the inherited theorem register.

ID	Grade	Scientific ruling
T1	Internal finite-order lemma	Unique order follows if the seven dependency relations form the asserted chain; the partial order and uniqueness must be written explicitly.
T2	Internal uniqueness lemma	The Fibonacci update is unique only within the declared class of nonnegative copy-and-add matrices.
T3	Exact algebra	The positive projective fixed point of the stated recurrence is φ . Physical relevance is separate.
T4	Exact conditional algebra	$r = (2\varphi)^{-1}$ follows from the A_3 normalization; the factor of two is assumed.
T5	Exact identity	$(u + r)^2 = u^2 + 2ur + r^2 = 1$. Sector names do not follow from the algebra.
T6	Conjectural quantization rule	Binary record status does not entail a physical crossing price exactly r . Preserve as a testable bridge.
T7	Exact conditional identity	Given r and three crossings, $r^3 = (\sqrt{5} - 2)/8$.
T8	Conditional classification	The rank-eight positive-definite even unimodular lattice is uniquely E_8 ; rank, integrality, evenness, unimodularity, and positivity must each be supplied (Conway and Sloane, 1999).
T9	Incomplete argument	Reading a relation twice does not prove an even integral bilinear form. This is an open target lemma.

ID	Grade	Scientific ruling
T10	Root-system lemma; conditional physics	Suitable E_8 roots close a triangle. Its status as the unique minimal physical history depends on admissible-move rules.
T11	Exact geometry, corrected	For roots of squared norm two with inner product -1 , parallelogram area is $\sqrt{3}$ and triangle area is $\sqrt{3}/2$.
T12	Reproducible computation	The Coxeter-plane radii exhibit the stated golden pairing under the recorded projection and tolerance; a symbolic proof would strengthen the result.
T13	Internal operator result	If $C^2 = C$, then $\ker(I - C) = \text{ran } C$. Calling the range physical is an identification.
T14	Exact operator observation	$I - C$ has a restricted spectrum; a general positive generator with the same kernel need not. No mass spectrum follows.
T15	Structural assumption	One-dimensional temporal order is not derived from the word “record”; rank-one time is an explicit premise.
T16	Structural assumption	One enacted closure does not prove a one-dimensional closure representation.
T17	Conditional metric construction	A declared three-plus-one split with opposite signs is Lorentzian. The ranks and signs require independent support.
T18	Established theorem, delimited	Alexandrov–Zeeman-type causal-preserver theorems recover Lorentz/conformal structure under dimensional, bijective, and causal hypotheses (Zeeman, 1964; Alexandrov, 1967).
T19	Withdrawn as proof	Lack of a distinguishing event does not force coherent quantum superposition; a classical mixture remains an alternative. A full operational reconstruction is required (Chiribella et al., 2011).
T20	Conditional representation	Continuous unit-modulus characters of additive action are exponentials. Complex Hilbert structure needs further composition and operational axioms.

ID	Grade	Scientific ruling
T21	Established theorem under hypotheses	Gleason/Busch/Caves-type results constrain noncontextual additive probability measures; fairness language alone is insufficient (Gleason, 1957; Busch, 2003; Caves et al., 2004).
T22	Established theorems under hypotheses	Wigner applies to transition-probability-preserving bijections and Stone to strongly continuous unitary groups; identifying the generator as energy is physical (Wigner, 1959; Stone, 1932).
T23	Exact algebra; conditional physics	The cross term is $2ur$. Universal interpretation as interference or boundary weight needs a model-specific amplitude map.
T24	Algebraic construction; conjectural gravity	A squared defect has three coefficient classes. A gravitational action also requires connection, curvature, invariant pairing, dimensions, symmetry breaking, and variation (MacDowell and Mansouri, 1977).
T25	Classification proposal	Commuting, sector-connecting, and noncommuting operations form a useful taxonomy; they do not derive gauge particles, matter, or masses.
T26	Persistence criterion	Retention rather than overwrite defines longitudinal identity within the framework; it is not a universal theorem forbidding replacement dynamics.
T27	Internal consequence plus interpretation	A coherent state is not a definite record under D11. Intrinsic unreadability of outcome selection remains interpretive.
T28	Physical conjecture	General relativity admits recollapsing cosmologies. A no-overwrite law must be realised covariantly before it can exclude them.
T29	Conditional rank bookkeeping	If the geometry block has rank five and time and closure each rank one, the remaining external rank is three.
T30	Finite Lie-theoretic result	The stated rank-three closed-root-subsystem classes may be retained with an exact classification or exhaustive certificate and a precise definition of subsystem.

ID	Grade	Scientific ruling
T31	Multi-stage conjectural selection	Complement calculations can be exact; irreducibility alone does not force one decay operator, and a 1 + 3 index split does not derive three particle families.
T32	Reproducible finite construction	Antipodal roots guarantee zero total sum; the claimed 240-step ordering additionally requires explicit adjacency conditions and an independently checked witness.
T33	Conjectural action identification	The coefficient pattern contacts MacDowell–Mansouri gravity, but term-level identity awaits the full differential-form action and field equations.
T34	Open; inherited proof rejected	One closure does not imply every loop contracts, and one world does not imply simple connectivity. Perelman’s theorem applies only after a closed simply connected three-manifold is established (Perelman, 2002).
T35	Established hydrogen mathematics; circular bridge	Fock’s S^3 follows from the Coulomb problem. Finite-design quadrature is valid, but importing that sphere cannot independently derive hydrogen (Fock, 1935; Delsarte et al., 1977).
T36	Conjectural chirality mechanism	A finite decay group alone does not provide a complete Lorentz-covariant chiral field theory or evade all E_8 representation constraints (Distler and Garibaldi, 2010).
T37	Conditional operator result plus finite search	Tomiyama and finite-group averaging apply under specified algebraic hypotheses; uniqueness and completeness of the root-generated search must be certified (Tomiyama, 1957).
T38	Standard GUT reproduction	The exterior-algebra charge table and 3/8 unification value are established $SU(5)/SO(10)$ representation theory. The traversal origin of the three-plus-two split remains unproved.
T39	Exact internal number; speculative map	The power of r is algebraically checkable. Its equality with the measured Planck/vacuum hierarchy requires a dimensionally complete, non-circular convention and uncertainty map.

4.1 Principal corrections

Several rulings materially change the inherited register.

1. T6 no longer treats binary record status as a derivation of an exact crossing cost.
2. T9 remains open because evenness requires an integral bilinear form, not a verbal two-sided reading.
3. T11 distinguishes the parallelogram area $\sqrt{3}$ from the triangular area $\sqrt{3}/2$.
4. T15 and T16 are structural rank assumptions, not proofs from singleness language.
5. T19's claimed derivation of superposition is withdrawn.
6. T28 is not a theorem of general relativity.
7. T34's route to S^3 is rejected because closure and uniqueness do not imply simple connectivity.
8. T35 retains the finite-design result but rejects the circular hydrogen derivation.
9. T36 does not use a finite group as a blanket escape from a field-theoretic no-go result.
10. T38 is credited to standard grand-unified representation theory.
11. T39 separates an exact internal number from its speculative physical map.

These are not cosmetic caveats. They determine what future work must calculate.

Reproducibility and computational certificates

5.1 Minimum reproducible unit

A numerical result is reproducible only when the following object exists:

$$\mathcal{U} = (I, C, E, O, V), \quad (5.1)$$

where I identifies immutable inputs, C identifies code, E records the execution environment, O contains outputs, and V independently verifies declared invariants. Each component must be content-addressed or versioned. A screenshot or rounded table is not a certificate.

For stochastic computation, I includes random seeds and the generator. For optimization or search, the certificate must separate discovery from verification. For exhaustive claims, the search space, pruning rules, symmetry quotient, and completion count must be explicit. For floating-point identities, tolerances and conditioning must be stated; symbolic or integer verification is preferred when available.

5.2 Certificate classes in the corpus

The reproducibility layer contains several distinct objects:

Algebra certificates

regenerate r, u , the three weights, closed forms, and cosmological dictionary values;

Root-system certificates

construct E_8 , projections, subgroup complements, ring radii, and candidate traversals;

Geometry certificates

expand the gravitational square and verify coefficient ratios;

Cosmology certificates

integrate background quantities, compute residual ledgers, and generate the master readout;

Data audits

verify the Atomic Web records and catalogue transformations;

Manifests

inventory paths and hashes but do not validate scientific conclusions.

5.3 Independent verification

The same implementation that discovers a witness should not be the sole verifier. For a root traversal, a small verifier should read the witness and check membership, uniqueness, adjacency, and closure without rerunning the search. For a cosmological table, a verifier should recompute formulas from a transparent parameter file and compare at full precision. For a document build, the source, bibliography, generated tables, compiler, and rendered pages are all part of the reproducible artefact.

EXPOSURE 6.1 — NUMERICAL PROOF BOUNDARIES

Machine-precision agreement is evidence for a computational statement, not automatically an exact theorem. An exhaustive result is only as complete as its enumeration and symmetry handling. The register must say whether a result is symbolic, integer-exact, interval-certified, exhaustively finite, or floating-point empirical.

5.4 Recommended release layout

Each public release should contain:

1. the five LaTeX sources and omnibus source;
2. the bibliography with persistent identifiers;
3. the full source and duplicate ledgers;
4. a machine-readable claim register;
5. data and code manifests with SHA-256 digests;
6. independent verification scripts;
7. build instructions and environment versions;
8. rendered PDFs plus page-image quality-control results;
9. a correction and supersession log.

Measurement, statistics, and falsification

6.1 The measurement dossier

Every empirical claim should ship with a dossier containing:

1. the physical mechanism and state variables;
2. the observable and its units;
3. the channel and propagation model;
4. instrument response, calibration, and selection function;
5. the estimator and full covariance;
6. nuisance parameters and priors;
7. the comparison model and likelihood;
8. preregistered acceptance, revision, and rejection rules;
9. data and code sufficient to reproduce the result where licensing permits.

This is the concrete implementation of the observational chain in Volume III.

6.2 Reconstruction is not prediction

The corpus contains several legitimate reconstructions: Standard Model charge branching, hydrogenic harmonic sampling, the MacDowell–Mansouri coefficient contact, and algebraic cosmological readouts. A reconstruction shows compatibility or compression. It becomes a prediction only if the model and analysis are fixed before the relevant target is used and if the result is sufficiently discriminating against alternatives.

For a dataset D , model M , nuisance parameters ν , and theory parameters θ , the comparison object is a likelihood or sampling distribution

$$p(D \mid \theta, \nu, M), \tag{6.1}$$

not a list of percentage differences. Correlated cosmological parameters require the joint covariance. A mean absolute pull calculated after selecting quantities

and transformations is descriptive unless selection is included in the inferential procedure.

6.3 Negative results belong in the main theory

The following negative results are scientifically productive and remain visible:

- vacuum crowding is too small to supply galactic dark matter in the tested form;
- geometry alone cannot separate α_L from ν in the Bubble compression kernel;
- family projections that give equal readouts do not explain observed family hierarchies;
- the Saha CMB solver places last scattering too early and does not calculate the observed spectra;
- the hydrogen sphere bridge is circular without an independently derived Coulomb interaction;
- the superposition, S^3 , chirality, and vacuum-hierarchy arguments do not presently reach their inherited grades;
- the cosmological master score is in-sample and does not possess a calibrated discovery significance.

6.4 Falsification must be model-specific

A framework this broad cannot be defeated by one vague mismatch, but each physical realisation can and should expose itself. A falsifier must bind a named version, observable map, parameter freedom, data domain, and threshold. It must also state what survives. Rejecting the dark-matter realisation would not invalidate the golden-ratio identity; rejecting the closure-to-e-fold map would not invalidate the finite root construction.

Historical consolidation and supersession

7.1 How the corpus developed

The source chronology can be summarized in five scientific phases:

1. **Foundational grammar:** distinction, identity, relation, the seven doings, the two-channel recurrence, and the three weights;
2. **Carrier construction:** E_8 , root geometry, clustering, decay, and the readable algebra;
3. **Traversal and observation:** helical lift, record order, quantum interfaces, Logical Action, Personal Relativity, and Universopedia;
4. **Physical applications:** gravity, particle branching, hydrogen, dark sectors, cosmological dictionaries, and empirical programmes;
5. **Audit and capstone:** formal registers, addenda, computational certificates, failure ledgers, the Field Ledger, and later matter/rate papers.

The five-volume architecture follows this dependency chronology rather than file creation time, because later documents frequently restate earlier material more clearly and occasionally correct it.

7.2 Consolidation rules

Exact duplicate files are represented once scientifically and many times in provenance. PDF and DOCX copies with the same intellectual content are treated as renderings of one source unless their text differs. Generated tables are not counted as independent evidence when they are deterministic outputs of the same script. Later restatements are merged into the strongest readable formulation while divergences enter the supersession ledger.

No information is discarded merely because it is scientifically weak. Instead it receives one of four dispositions:

integrated, historical, superseded, withdrawn. (7.1)

This prevents both inflation of the series by repetition and erasure of the research history.

7.3 Named supersessions

The principal supersessions include:

- “traversal payment as a universal one-third constant” becomes an audit-dependent typing score;
- “superposition forced by nonselection” becomes an operational postulate awaiting reconstruction;
- “three modes plus closure imply S^3 ” becomes an open topology problem;
- “hydrogen derived from carrier self-duality” becomes a finite-design quadrature result plus an open interaction bridge;
- “MacDowell–Mansouri term for term” becomes a coefficient contact awaiting a complete action;
- “finite decay evades the E_8 chirality obstruction” becomes an unproved field-theory programme;
- “vacuum crowding supplies galactic dark matter” is withdrawn;
- every edition freeze becomes a historical analysis lock, not an immutable scientific state.

Coverage assurance and handoff

8.1 What complete coverage means

Complete coverage is an auditable relation between corpus sources and series dispositions, not an assertion that every duplicated paragraph appears verbatim. The crosswalk in the final chapter lists all 491 unique text-bearing sources. The companion CSV retains the complete fields and is the authoritative audit table. The full 1,319-file inventory and duplicate groups preserve non-text assets and alternate copies.

Coverage is checked at three levels:

Source level

every unique source has a volume, section, role, and disposition;

Claim level

each substantive mathematical or physical claim is integrated, regraded, superseded, withdrawn, or named as open;

Artefact level

code, data, certificates, images, manifests, and rendered copies remain traceable even when not printed in prose.

8.2 Open programme in dependency order

The shortest scientifically meaningful route forward is:

1. formalize the distinction/record axioms and remove arguments that depend only on word choice;
2. state and certify the exact lattice and subgroup hypotheses;
3. construct the local carrier/cluster action and derive field equations;
4. produce an anomaly-free chiral matter sector with masses and running couplings;
5. derive dark-sector stress–energy and cosmological perturbations;
6. replace the Saha approximation with full recombination and Boltzmann evolution;

7. derive prospective observable maps for closure count, Hubble basins, vacuum response, and short-range gravity;
8. preregister and run experiments or withheld-data tests;
9. revise the living register regardless of whether the result is favorable.

8.3 Final scientific position

Unified Mechanics is presently a structured research programme containing exact algebra, nontrivial finite geometry, established mathematical contacts, reproducible computations, interpretive principles, and ambitious physical identifications. It is not yet a closed fundamental theory. Its strongest scientific form is the one adopted by this series: retain the exact results, credit established antecedents, state every load-bearing assumption, expose every missing bridge, preserve failures, and make the next calculations capable of changing the theory.

Complete source-disposition crosswalk

This appendix enumerates every unique text-bearing source in the indexed corpus: 491 records after exact-hash deduplication. The authoritative machine-readable ledger retains the absolute path, full SHA-256 digest, format, topic labels, claim labels, planned section, and audit status. The compact table below prints the fields required to trace every source into the series.

TABLE 9.1. Crosswalk population by source role.

Role	Unique sources
data or certificate	105
historical manuscript	6
implementation	148
manifest	19
manuscript	126
orientation or ledger	23
supporting text	64
Total	491

Volume assignments may contain more than one Roman numeral where a source supports several stages of the argument. Role codes are MS, HIST, CODE, CERT, NOTE, LEDGER, and MANIFEST, corresponding to the full roles above. A source listed here is not thereby endorsed; its disposition states where it was preserved, consolidated, corrected, superseded, or used as a certificate.

TABLE 9.2. Complete crosswalk of unique text-bearing corpus sources.

ID	Volume	Role	Source and disposition
S0001	I	NOTE	LEGACY_UNIFIED_MECHANICS/00_DERIVATION.md
		da9ce959873f	<i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
S0002	I	NOTE	LEGACY_UNIFIED_MECHANICS/00_LAWS.md

Continued on the following page

Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0003	IV	NOTE	b595ab15262a <i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/00_MASTER_HANDOVER .md
S0004	I	NOTE	2coc851473a4 <i>Cosmology and phenomenology; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/00_WHAT_THIS_IS .md
S0005	I	NOTE	34d5cd73a6e6 <i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/01_FOUNDATION .md
S0006	IV; V	NOTE	9o85f88cfcfd <i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/02_COSMOLOGY .md
S0007	II; V	NOTE	o64c67272c8b <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/03_GRAVITY_AND_BLACK_HOLES .md
S0008	IV; V	NOTE	8d17833c2bba <i>Legacy physical claims and supersession audit; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/04_QUANTUM_AND_HOLOGRAPHIC .md
S0009	II; V	NOTE	79c1f9c79fb8 <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/05_PARTICLE_PHYSICS .md
S0010	II; V	NOTE	8747485aa278 <i>Legacy physical claims and supersession audit; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/06_HETEROTIC_IDENTIFICATION .md
S0011	IV; V	NOTE	b9de92ffd493 <i>Legacy physical claims and supersession audit; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/07_EXPERIMENTAL_PROGRAM .md
S0012	IV; V	NOTE	cf5da971bo1e <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/08_EMPIRICAL_VALIDATION .md
S0013	III; V	NOTE	o292dba1f6e7 <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/11_DYNAMICS .md
			82c5959f866f <i>Legacy dynamics and supersession audit; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0014	III; V	NOTE	LEGACY_UNIFIED_MECHANICS/12_ARROW_OF_TIME.md 23da86dobaf6 <i>Legacy dynamics and supersession audit; mapped provisional</i>
S0015	III; V	NOTE	LEGACY_UNIFIED_MECHANICS/13_QUANTIZATION.md 963395e32a12 <i>Legacy dynamics and supersession audit; mapped provisional</i>
S0016	IV; V	NOTE	LEGACY_UNIFIED_MECHANICS/14_THERMODYNAMICS.md d811e5ac7dbf <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
S0017	IV; V	NOTE	LEGACY_UNIFIED_MECHANICS/15_STATISTICAL_MECHANICS.md 784e4169aeac <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
S0018	IV; V	NOTE	LEGACY_UNIFIED_MECHANICS/16_ELECTROMAGNETISM.md d8570ba4ec10 <i>Legacy phenomenology and scientific regrading; mapped provisional</i>
S0019	IV	NOTE	LEGACY_UNIFIED_MECHANICS/FAQ.md 1b6876bcbaad <i>Cosmology and phenomenology; mapped provisional</i>
S0020	V	NOTE	LEGACY_UNIFIED_MECHANICS/foundations/Canonicalization_Complete_Edition_1.md f891d92e527c <i>Methods and reproducibility; mapped provisional</i>
S0021	V	LEDGER	LEGACY_UNIFIED_MECHANICS/foundations/README.md 72595cc14f59 <i>Legacy corpus provenance; mapped provisional</i>
S0022	II	MS	LEGACY_UNIFIED_MECHANICS/Gravitational_Memory.pdf db106cb9ffie <i>Geometry and gravitation; mapped provisional</i>
S0023	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/__init__.py 24b91b8738b8 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0024	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/camb_backend.py 388o83282a77 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0025	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/channels/__init__.py a7f76b1855aa <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0026	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/channels/ barrier.py 095cde81840c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0027	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/channels/ light.py a3ad1a0ec80e <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0028	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/channels/ matter.py 119fd67846f2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0029	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/ cosmology.py b7ec4ccc7716 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0030	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/ derivations/__init__.py 5b28c2651doc <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0031	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/ derivations/lcdm_inputs.py e7dafa5c93f3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0032	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/lcdm.py 6212bd30a475 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0033	V	CODE	LEGACY_UNIFIED_MECHANICS/limb/um.py 3066a29d8d85 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0034	III	MS	LEGACY_UNIFIED_MECHANICS/papers/alpha_ em_g2.pdf 0dac2e2b8ab1 <i>Traversal and retained history; mapped provisional</i>
S0035	III	NOTE	LEGACY_UNIFIED_MECHANICS/papers/alpha_ em_g2.tex 62bad4bb766a <i>Traversal and retained history; mapped provisional</i>
S0036	II	MS	LEGACY_UNIFIED_MECHANICS/papers/ckm_ mixing.pdf 732ae2c56321 <i>Geometry and gravitation; mapped provisional</i>
S0037	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/ckm_ mixing.tex 16c559e00ba3 <i>Cosmology and phenomenology; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0038	IV	MS	LEGACY_UNIFIED_MECHANICS/papers/cosm_constant.pdf 92bo6dc46199 <i>Cosmology and phenomenology; mapped provisional</i>
S0039	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/cosm_constant.tex 716eaa40eb3f <i>Cosmology and phenomenology; mapped provisional</i>
S0040	IV	MS	LEGACY_UNIFIED_MECHANICS/papers/electroweak_baryon.pdf a0af7fd6554 <i>Matter and atomic applications; mapped provisional</i>
S0041	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/electroweak_baryon.tex 53f776957c54 <i>Matter and atomic applications; mapped provisional</i>
S0042	IV	MS	LEGACY_UNIFIED_MECHANICS/papers/extended/addendum_closed_results.pdf 99a955287489 <i>Cosmology and phenomenology; mapped provisional</i>
S0043	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/extended/addendum_closed_results.tex d62b2f652d77 <i>Cosmology and phenomenology; mapped provisional</i>
S0044	IV	MS	LEGACY_UNIFIED_MECHANICS/papers/gauge_unification.pdf 13do6e67d47c <i>Cosmology and phenomenology; mapped provisional</i>
S0045	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/gauge_unification.tex de5a6fd33c6b <i>Cosmology and phenomenology; mapped provisional</i>
S0046	IV	MS	LEGACY_UNIFIED_MECHANICS/papers/matter_antimatter.pdf fd1e84ff8428 <i>Cosmology and phenomenology; mapped provisional</i>
S0047	IV	NOTE	LEGACY_UNIFIED_MECHANICS/papers/matter_antimatter.tex fd5779b6bf71 <i>Cosmology and phenomenology; mapped provisional</i>
S0048	III	MS	LEGACY_UNIFIED_MECHANICS/papers/muon_g2.pdf fd314358796e <i>Traversal and retained history; mapped provisional</i>
S0049	III	NOTE	LEGACY_UNIFIED_MECHANICS/papers/muon_g2.tex bc577785caza <i>Traversal and retained history; mapped provisional</i>
S0050	III	MS	LEGACY_UNIFIED_MECHANICS/papers/neutrino_masses.pdf

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ID	Volume	Role	Source and disposition
S0051	III	NOTE	159f77633795 <i>Traversal and retained history; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/papers/ neutrino_masses.tex
S0052	III	MS	8ee6562bb201 <i>Traversal and retained history; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/papers/qcd_ running.pdf
S0053	III	NOTE	fcdf86cf22be <i>Traversal and retained history; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/papers/qcd_ running.tex
S0054	II	MS	17d2381ede10 <i>Traversal and retained history; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/papers/shared_ space.pdf
S0055	II	NOTE	3f762f9cbb56 <i>Geometry and gravitation; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/papers/shared_ space.tex
S0056	IV	NOTE	f88d95984b98 <i>Geometry and gravitation; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/PRE_ REGISTRATION.md
S0057	IV	MS	8c3085fo22c9 <i>Cosmology and phenomenology; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/Quantum Mechanics and Big Bang.pdf
S0058	V	LEDGER	05ao31e61ba5 <i>Cosmology and phenomenology; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/README.md
S0059	V	CODE	83a75f1bdfa5 <i>Legacy corpus provenance; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/research/ alpha_em_g2.py
S0060	V	CODE	148c77bf89c6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/research/ckm_ mixing.py
S0061	V	CODE	fa9be9f24fe7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/research/cosm_ constant.py
S0062	IV	NOTE	558ad7ffa191 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_B.yaml
			eb487ff15fa1 <i>Cosmology and phenomenology; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0063	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_L0.yaml 850123deac86 <i>Cosmology and phenomenology; mapped provisional</i>
S0064	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_L1.yaml 94598a4ecd56 <i>Cosmology and phenomenology; mapped provisional</i>
S0065	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_LF.yaml aa9006001469 <i>Cosmology and phenomenology; mapped provisional</i>
S0066	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_LFP.yaml 7beabaz86422 <i>Cosmology and phenomenology; mapped provisional</i>
S0067	V	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_M.yaml 58e42573f43b <i>Methods and reproducibility; mapped provisional</i>
S0068	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_R5v8.yaml 1e28da79e809 <i>Cosmology and phenomenology; mapped provisional</i>
S0069	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_T.yaml 3ec828b225ef <i>Cosmology and phenomenology; mapped provisional</i>
S0070	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ chain_yamls/chain_T2.yaml 85a0737293do <i>Cosmology and phenomenology; mapped provisional</i>
S0071	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/dgf/ PROGRAMME_PAPER.md 07cf6454d957 <i>Cosmology and phenomenology; mapped provisional</i>
S0072	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ gauge_unification.py 223459co81ao <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0073	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ matter_antimatter.py 61oad7eb3ob <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0074	V	CODE	LEGACY_UNIFIED_MECHANICS/research/muon_ g2.py 09f6ffafid6b <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0075	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ neutrino_masses.py f49528313638 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0076	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ particle_constants.py 18dda42341bc <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0077	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/02_gaia_targets/cmb_gstar_ targets.json 974e2e8b46e2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0078	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/02_gaia_targets/cmb_ranked_ targets.json bad8f85ce60c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0079	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/04_skypy_lss/skypy_highl_ pipeline.py 9515ce939283 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0080	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/04_skypy_lss/skypy_lss_ pipeline.py 6f13e3160a51 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0081	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/05_catalogue/ exoplanets.json 3983e9b28af3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0082	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/05_catalogue/MASTER_ CATALOGUE.md f9bcdae13415 <i>Cosmology and phenomenology; mapped provisional</i>
S0083	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/06_planck/find_planck_ seed.py a5858afbaa0c <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0084	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/06_planck/planck_real_ targeting.py 25442c161291 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0085	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/06_planck/planck_render.py d5ee819452d0 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0086	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/07_planet_concept/ OUROBOROS_SYSTEM.md d63836c8354a <i>Cosmology and phenomenology; mapped provisional</i>
S0087	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/07_planet_concept/render_ planet.py o6d57b205e60 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0088	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/07_planet_concept/render_ system_map.py de3d312f99a2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0089	II	NOTE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/EMPIRICAL_ EVIDENCE.md 5f4236fco7be <i>Geometry and gravitation; mapped provisional</i>
S0090	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/evidence_ summary.json 33f985b26c87 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0091	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/gather_ evidence.py 39c9e9d48ce7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0092	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/gather_ evidence2.py 7e5324ee025d <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0093	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/gather_ evidence3.py a7c6dz7f114b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0094	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/08_evidence/tess_plot_ bls.py b9393c6e5656 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0095	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/09_age_match/age_match.py dc3e3oac2ocf <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0096	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/09_age_match/age_matched_ catalogue.json 3ddoc419a254 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0097	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/09_age_match/OPS_ CATALOGUE.md 88fdo41465e1 <i>Cosmology and phenomenology; mapped provisional</i>
S0098	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/09_age_match/ops_ targets.json fo37d8be2fa4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0099	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/10_earth_ages/bio_peer_ rank.py e956a7c75300 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0100	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/10_earth_ages/bio_peer_ ranked.json o9bf60289095 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0101	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/10_earth_ages/earth_ages.py do2464ef3f76 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0102	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/10_earth_ages/ouroboros_ matter_age.py a566a2a860ea <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0103	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/11_dark_sector/dark_sector_ geometry.py 6f21851c0376 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0104	IV	NOTE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/ouroboros/LIGHT_MATTER_ AGE.md d3670bfe2168 <i>Cosmology and phenomenology; mapped provisional</i>
S0105	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/ouroboros/simulate_ ouroboros.py 0a86ca9e810a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0106	V	LEDGER	LEGACY_UNIFIED_MECHANICS/research/ planet_hunt/README.md 33c3f5289f27 <i>Legacy corpus provenance; mapped provisional</i>
S0107	V	CODE	LEGACY_UNIFIED_MECHANICS/research/qcd_ running.py 44b51a79ef9c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0108	V	LEDGER	LEGACY_UNIFIED_MECHANICS/research/ recursion_floor/README.md c41eaa23be47 <i>Legacy corpus provenance; mapped provisional</i>
S0109	V	CERT	LEGACY_UNIFIED_MECHANICS/research/ recursion_floor/runA_recursion_ study.json 330dfa13ebee <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0110	V	CODE	LEGACY_UNIFIED_MECHANICS/research/ render_cmb_4k.py 02c2fd4ec8a2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0111	V	CODE	LEGACY_UNIFIED_MECHANICS/research/um_ derived_constants.py 70fa49a142e4 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0112	I	NOTE	LEGACY_UNIFIED_MECHANICS/START_HERE.md b16c4669bbf7 <i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
S0113	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ alternate_recursions.json od3b517203d1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0114	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ cereal_bowl_weights.json 99a57d21fe09 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0115	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ chains/_summary.json dd367cbfebd1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0116	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ chains/_summary_v2.json 5708e140649b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0117	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ demo_chain/demo_chain_summary.json d58a88be37b3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0118	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ everything_suite.json 53d010da7cae <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0119	IV	NOTE	LEGACY_UNIFIED_MECHANICS/tests/data/ everything_suite_table.md ae907c1bc7cf <i>Cosmology and phenomenology; mapped provisional</i>
S0120	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ million_test_summary.json 525da799a744 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0121	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ nautilus_floor.json 4e991aada1c4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0122	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/ nautilus_published.json 44c6869ba96b <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0123	V	CERT	LEGACY_UNIFIED_MECHANICS/tests/data/structural_bayes.json 3ef3d461offd <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0124	V	CODE	LEGACY_UNIFIED_MECHANICS/tests/demo_chain.py affc970f2275 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0125	V	CODE	LEGACY_UNIFIED_MECHANICS/tests/test_limb_derivations.py d4f11745e368 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0126	I	NOTE	LEGACY_UNIFIED_MECHANICS/THE_QUESTION.md 440f6ca1ec01 <i>Legacy foundations regraded against the current formal chain; mapped provisional</i>
S0127	V	MS	MONOGRAPH/00 START HERE/00 The Corpus Atlas.docx eafbb575e416 <i>Methods and reproducibility; mapped provisional</i>
S0128	V	MS	MONOGRAPH/00 START HERE/00 The Corpus Atlas.pdf 2829e4d73fo4 <i>Methods and reproducibility; mapped provisional</i>
S0129	V	LEDGER	MONOGRAPH/00 START HERE/01 README.md 2ce5b096fe39 <i>Methods and reproducibility; mapped provisional</i>
S0130	V	LEDGER	MONOGRAPH/00 START HERE/02 Edition Note – Fifth Printing.md 0b5621b84779 <i>Methods and reproducibility; mapped provisional</i>
S0131	V	MANIFEST	MONOGRAPH/00 START HERE/03 CORE FREEZE MANIFEST.md 6561cd10cb60 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0132	V	MANIFEST	MONOGRAPH/00 START HERE/04 SHA256SUMS.txt 187691c436e9 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0133	V	MS	MONOGRAPH/01 CORE BOOKS/A – A World of Distinctions.pdf 9889d4e736fd <i>Cross-format verification copy; mapped provisional</i>
S0134	V	MS	MONOGRAPH/01 CORE BOOKS/B – Universopedia, Combined Reading Edition.pdf

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ID	Volume	Role	Source and disposition
S0135	III	MS	e2e155d38733 <i>Cross-format verification copy; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/B1 – Universopedia, The Observational Standard.pdf
S0136	III	MS	6b87313041c2 <i>Observer resolution and readout; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/B2 – Universopedia, Systems of Interpretation.pdf
S0137	V	MS	6af28c33316b <i>Observer resolution and readout; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/C – Unified Mechanics, The Formal Register.pdf
S0138	I	MS	b5f8c8efid1a <i>Cross-format verification copy; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/Sources/A – A World of Distinctions.docx
S0139	IV	MS	4e611ce338ef <i>Continuous foundations narrative; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/Sources/B1 – Universopedia, The Observational Standard.docx
S0140	IV	MS	3bodb48fcbfb <i>Observational contract and cross-scale ledger; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/Sources/B2 – Universopedia, Systems of Interpretation.docx
S0141	V	MS	bc5b5bdfbo35 <i>Applications and negative screens; mapped provisional</i>
			MONOGRAPH/01 CORE BOOKS/Sources/C – Unified Mechanics, The Formal Register.docx
S0142	V	MS	3d54odco1d25 <i>Formal register and dependency graph; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/00 The Papers Index.docx
S0143	V	MS	891bd6933957 <i>Paper sequence and provenance; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/00 The Papers Index.pdf
S0144	I	MS	7b8268fdo686 <i>Paper sequence and provenance; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/04 The Realisation and Self-Description Principle.docx
			d661accc34ba <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0145	I	MS	MONOGRAPH/02 TECHNICAL PAPERS/04 The Realisation and Self-Description Principle.pdf 7eea3cb72b9f <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
S0146	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/05 The Snowflake Realisation Closure Theorem.docx d03f9de393c8 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0147	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/05 The Snowflake Realisation Closure Theorem.pdf 65c3f2cefd95 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0148	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/08 Research Notes – The Self-Dual Carrier and the Silver Interface.docx 2558ac874ef6 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0149	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/08 Research Notes – The Self-Dual Carrier and the Silver Interface.pdf 9a0f49131128 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0150	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/09 Interpretations – The Golden Skeleton of E8.docx 304a50cd7ceo <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0151	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/09 Interpretations – The Golden Skeleton of E8.pdf de520f6f8af2 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0152	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/10 The Interface Work Integral – Settling the Root Three.docx bc205825529e <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0153	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/10 The Interface Work Integral – Settling the Root Three.pdf

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ID	Volume	Role	Source and disposition
S0154	II	MS	9e89058a3ec2 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/11 End-to-End Interface Closure – The Theorem Queue.docx
S0155	II	MS	3f69cacf44e5 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/11 End-to-End Interface Closure – The Theorem Queue.pdf
S0156	II	MS	39296288ab21 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/12 Theorem Queue Proof Ledger.docx
S0157	II	MS	3743ffa544ac <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/12 Theorem Queue Proof Ledger.pdf
S0158	I	MS	7dbf50066ba5 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/13 The Logic of Physical Action (issued as Monograph II).docx
S0159	I	MS	73a67259e18d <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/13 The Logic of Physical Action (issued as Monograph II).pdf
S0160	I	MS	dde68f7a96eo <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/14 Before the Form – Spencer-Brown and Traversal.docx
S0161	I	MS	cd82coc1b1de <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/14 Before the Form – Spencer-Brown and Traversal.pdf
S0162	I	MS	c9fa7113eb59 <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
			MONOGRAPH/02 TECHNICAL PAPERS/15 Bounded Possibility.docx
			48299c509c82 <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0163	I	MS	MONOGRAPH/02 TECHNICAL PAPERS/15 Bounded Possibility.pdf 79699b231ae2 <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
S0164	I; III	MS	MONOGRAPH/02 TECHNICAL PAPERS/16 Principles of Observation and Mechanical Observation.docx fe30a24ffe4 <i>Observation and mechanical readout; mapped provisional</i>
S0165	I; III	MS	MONOGRAPH/02 TECHNICAL PAPERS/16 Principles of Observation and Mechanical Observation.pdf 5c8c2d2a61f2 <i>Observation and mechanical readout; mapped provisional</i>
S0166	I	MS	MONOGRAPH/02 TECHNICAL PAPERS/ 17 Field Assembly and Lossless Transcription.docx d5748ff4b67f <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
S0167	I	MS	MONOGRAPH/02 TECHNICAL PAPERS/17 Field Assembly and Lossless Transcription.pdf 69a44f277d99 <i>Logical action, systemhood, and lossless transcription; mapped provisional</i>
S0168	III	MS	MONOGRAPH/02 TECHNICAL PAPERS/18 The Quantised Crossing and the Closure of the Open Ledger.docx odf516ae9993 <i>Quantised crossing, records, and closing ledger; mapped provisional</i>
S0169	III	MS	MONOGRAPH/02 TECHNICAL PAPERS/18 The Quantised Crossing and the Closure of the Open Ledger.pdf 9375055578d2 <i>Quantised crossing, records, and closing ledger; mapped provisional</i>
S0170	III	MS	MONOGRAPH/02 TECHNICAL PAPERS/19 The Closing Ledger.docx abce6c244498 <i>Quantised crossing, records, and closing ledger; mapped provisional</i>
S0171	III	MS	MONOGRAPH/02 TECHNICAL PAPERS/19 The Closing Ledger.pdf d94af56fa81b <i>Quantised crossing, records, and closing ledger; mapped provisional</i>
S0172	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/20 The Forward Closure.docx 987205b7043d <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0173	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/20 The Forward Closure.pdf 9c24c95a4ed7 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0174	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/21 The Rank-Three Decay.docx c287003bce21 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0175	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/21 The Rank-Three Decay.pdf 11dc724074c9 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0176	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/22 Chirality from the Arrow.docx e869d362eb34 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0177	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/22 Chirality from the Arrow.pdf 26c8ec7f66cf <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0178	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/23 The Design Strength of the Carrier.docx 019b4b12c7cb <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0179	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/23 The Design Strength of the Carrier.pdf be1812e5908c <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0180	II; IV; V	MS	MONOGRAPH/02 TECHNICAL PAPERS/24 What It Now Claims, and What Would End It.docx 320b76deb620 <i>Capstone claims, phenomenology, and defeat conditions; mapped provisional</i>
S0181	II; IV; V	MS	MONOGRAPH/02 TECHNICAL PAPERS/24 What It Now Claims, and What Would End It.pdf 39c9ddd32b6b <i>Capstone claims, phenomenology, and defeat conditions; mapped provisional</i>
S0182	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/25 The Matter Action.docx b562955f2ce2 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i>
S0183	II	MS	MONOGRAPH/02 TECHNICAL PAPERS/25 The Matter Action.pdf

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ID	Volume	Role	Source and disposition
S0184	II	MS	a26ecf19c617 <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i> MONOGRAPH/02 TECHNICAL PAPERS/26 Forcing the Rest.docx
			a8c5dc44bb3c <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i> MONOGRAPH/02 TECHNICAL PAPERS/26 Forcing the Rest.pdf
S0185	II	MS	728098cccodz <i>Carrier, reduction, geometry, and physical sectors; mapped provisional</i> MONOGRAPH/03 COMPANION/Companion 1 – The Semantic Word System.pdf
			15126a66e437 <i>Logical action and system foundations; mapped provisional</i> MONOGRAPH/03 COMPANION/Companion 2 – The Snowflake Observatory (interactive).html
S0186	I	MS	a769bf38765c <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/03 COMPANION/Companion 3 – A Sentence Describing Itself (interactive).html
			acebf72e39ef <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/03 COMPANION/Companion 4 – A World of Distinctions (interactive monograph).html
S0187	V	CODE	cb95cbf2860z <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/03 COMPANION/The Bubble Field/00 STANDING ACCOUNT.md
			f4aa93eb323d <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i> MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/15_the_bubble_ field.json
S0188	V	CODE	34b66873fooc <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/audit_closure_ attack_results.json
			4757f38f65b6 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0193	V	CODE	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/bubble_field.py e38d3e3fc205 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0194	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/bubble_field_closure_attack_results.json 2ba53bbcc230 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0195	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/bubble_field_results.json 7foc5050aee9 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0196	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/bubble_field_rigorous_results.json 3758b09d5b96 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0197	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/exercising_the_field_results.json d202078c4715 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0198	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/three_part_bubble_results.json fo4ce30b1a1b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0199	V	CERT	MONOGRAPH/03 COMPANION/The Bubble Field/01 Computations/verify_review_results.json d2699a5c802c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0200	V	CODE	MONOGRAPH/03 COMPANION/The Bubble Field/02 Interactive/universal-field-exoplanet-lab.html 7698af2fi7e1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0201	V	CODE	MONOGRAPH/03 COMPANION/The Bubble Field/02 Interactive/universal-field-solar-system.html

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ID	Volume	Role	Source and disposition
S0202	III	MS	82adof238e7f <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 01 – The Realisation Compiler (Stellar Carbon- Oxygen Example).pdf
S0203	III	MS	58cef29bba58 <i>Observer resolution and readout; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 02 – The Processing-Machine Motif and a Neuronal Test.docx
S0204	III	MS	c14fdc6ebc3d <i>Observer resolution and readout; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 02 – The Processing-Machine Motif and a Neuronal Test.pdf
S0205	II	MS	73bccee62a49 <i>Observer resolution and readout; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 03 – The Accumulating-History Extraction (Cortical Neurogenesis).docx
S0206	II	MS	c21a8aoe1dcc <i>Least-action field; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 03 – The Accumulating-History Extraction (Cortical Neurogenesis).pdf
S0207	II	MS	251ab455edde <i>Least-action field; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 04 – The Balanced-Turnover Test (Subgranular Zone).docx
S0208	II	MS	dodo4f86d54d <i>Least-action field; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ application notes/Application 04 – The Balanced-Turnover Test (Subgranular Zone).pdf
S0209	V	CERT	e532b9d926e8 <i>Least-action field; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/chirality_certificate.json

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ID	Volume	Role	Source and disposition
S0210	V	CERT	3af0e340be4d <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/design_strength_ certificate.json
S0211	V	CERT	fc1eff3eb26e <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/families_certificate.json
S0212	V	CERT	3f85ec8e2bfe <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/forcing_certificate.json
S0213	V	CERT	fa55480ab1ao <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/graph_certificate.json
S0214	V	CERT	d3c331c0959b <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/hydrogen_certificate.json
S0215	V	CERT	d13d53ef7f91 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/lagrangian_ certificate.json
S0216	V	CERT	8ef31dfdfdd6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/matter_action_ certificate.json
S0217	V	CERT	4eec0ee05c98 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/o1_closure_ certificate.json
S0218	V	CERT	e3d7599e033b <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/o1_o3_certificate.json
S0219	V	CERT	65c21fe0db5c <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/04 REPRODUCIBILITY/ certificates/o2_certificate.json

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ID	Volume	Role	Source and disposition
S0220	V	CERT	161e446f759b <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/o2_elimination_ certificate.json
S0221	V	CERT	10a7b54c267a <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/o4_certificate.json
S0222	V	CERT	7fec939c348e <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/o4_closure_witness.json
S0223	V	CERT	269989053714 <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/o5_o6_certificate.json
S0224	V	CERT	57a3017961dd <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/o6_dark_energy_ certificate.json
S0225	V	CERT	4521bcd4de7 <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ certificates/run_manifest.json
S0226	V	CERT	2fe982714a87 <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/data_ manifest.csv
S0227	V	CERT	18ae375a3888 <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ dependencies.csv
S0228	V	CERT	7af88e77c536 <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/nodes.csv
S0229	V	NOTE	7eb191e7322a <i>Reproducibility assets and computational provenance; mapped provisional</i> MONOGRAPH/04 REPRODUCIBILITY/ requirements.txt
			1e42ae5551a9 <i>Methods and reproducibility; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0230	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ dev_cross.json 0a6e02f6cb60 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0231	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ dev_result.json 71e255d28fa1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0232	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ interface_results.json f5f9f844b5d3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0233	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ neuro_result.json 6e63a1fe7a77 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0234	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ sgz_result.json e6a67fb401e2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0235	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/results/ sparc_results.json ofbebbfee125 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0236	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/run_all.py 70fiba3471ae <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0237	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ chirality.py ab8f33fc8cd7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0238	V	CERT	MONOGRAPH/04 REPRODUCIBILITY/scripts/ chirality_certificate.json 8af8883b5d69 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0239	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ design_strength.py a85a8914b253 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0240	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ dev_neurogenesis_test.py c2d04cd7b8e2 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0241	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ distance_from_the_line.py d2a4fd17d5b5 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0242	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ e8_golden_tests.py b58b8502ccc2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0243	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ families.py 0181of9be0ee <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0244	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ forcing_the_rest.py 82ca09666f8c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0245	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ hydrogen_closure.py eb8f51916d99 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0246	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ interface_integral_test.py 256cf4457e26 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0247	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ lagrangian.py 2e6267fce3fd <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0248	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ matter_action.py 1e0cc1d54c57 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0249	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ neuro_avalanche_test.py 152be0f6bc91 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0250	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o1_closure.py 72d43905e3d0 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0251	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o1_o3_representation.py 79df9eea7ca4 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0252	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o2_elimination.py 8c9ca1537f3c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0253	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o2_surviving_block.py f3eee84bcf14 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0254	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o4_carrier_closure.py od24c463fc89 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0255	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o5_o6_cosmology.py 1d99d934284f <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0256	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ o6_dark_energy_audit.py a12c5368bef8 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0257	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ sgz_balanced_turnover_test.py 2bfd2586b391 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0258	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ sparc_test.py o4ca5598352e <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0259	V	CODE	MONOGRAPH/04 REPRODUCIBILITY/scripts/ validate_dependencies.py c429e4dd9bab <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0260	III	MS	MONOGRAPH/05 TRAVERSAL PAPERS/00 The Retained Fraction and the Discarded Half.docx 79db23a673b6 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
S0261	III	MS	MONOGRAPH/05 TRAVERSAL PAPERS/00 The Retained Fraction and the Discarded Half.pdf 4d83ao8442d5 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
S0262	III	MS	MONOGRAPH/05 TRAVERSAL PAPERS/01 The Depth of Traversal (Consolidated).docx

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0263	III	MS	c3cd55df595f <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/01 The Depth of Traversal (Consolidated).pdf
S0264	III	MS	ceaf95e27b49 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/02 The Rate Across the Scale.docx
S0265	III	MS	dbbff3bf2383 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/02 The Rate Across the Scale.pdf
S0266	III	MS	7a7212b8b5e2 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/03 From Chiral Wave Evolution to the Principle of Logical Action.docx
S0267	III	MS	63bc9eb09444 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/03 From Chiral Wave Evolution to the Principle of Logical Action.pdf
S0268	III	MS	86cbdba9016d <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/04 Least-Cost Holonomy and Co-Distinction Fields.docx
S0269	III	MS	04c31ao87780 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/04 Least-Cost Holonomy and Co-Distinction Fields.pdf
S0270	II	MS	2e3956593058 <i>Traversal, time, holonomy, and readout; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/05 The Source Sector.docx
S0271	II	MS	1dc635ea1167 <i>Source sector and gravitational coupling; mapped provisional</i>
			MONOGRAPH/05 TRAVERSAL PAPERS/05 The Source Sector.pdf
S0272	II	LEDGER	cb608005ec8f <i>Source sector and gravitational coupling; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/00 START HERE/01 README.md
S0273	II	LEDGER	f6abf5322ac7 <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/00 START HERE/02 STANDING ACCOUNT.md

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0274	II	LEDGER	9d8ba77fac84 <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/00 START HERE/03 OPEN LEDGER ADDITIONS.md
S0275	V	MANIFEST	7acobae35c3f <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/00 START HERE/04 SHA256SUMS.txt
S0276	II	LEDGER	f8ae94237899 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/00 START HERE/05 LEAN BUILD NOTE.md
S0277	II	MS	639824ff285e <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/ 01 THE FIELD/Geometric_Lift_of_ Negative_Holographic_Evolution_Joseph_ Shields.docx
S0278	II	NOTE	528803ede770 <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/01 THE FIELD/The Bubble Field(1).md
S0279	II	MS	4d4c993918bo <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/01 THE FIELD/The_Bubble_Field_Mathematical_ Field_Edition_Joseph_Shields.docx
S0280	II	MS	ecob4b78e23c <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/01 THE FIELD/Universal_Field_of_Least_Action_ Complete_Programme_Joseph_Shields.docx
S0281	V	CODE	6aa342f52f2f <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/audit_closure_attack.py
S0282	V	CERT	aad79524d925 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/audit_closure_attack_ results.json
			b5d7480d40c8 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0283	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field.py cf2e3c3fb2c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0284	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_closure_ attack.py ofd213ef2b58 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0285	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_closure_ attack_results.json 37cfae7f669 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0286	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_compression_ boundary.py 4387ccb3a925 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0287	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_compression_ boundary_results.json a094545536dd <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0288	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_results.json 9b251287a266 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0289	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_rigorous.py b1d99021645b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0290	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/bubble_field_rigorous_ results.json 3d3886c903bc <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0291	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/colour_field_closure.py 8e57ca1457ce <i>Reproducibility assets and computational provenance; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0292	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/colour_field_closure_ results.json dazbc849ebb7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0293	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/exercising_the_field.py eeca9oodb45f <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0294	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/exercising_the_field_ results.json o7ob4538a616 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0295	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/mass_operator.py 76e85816eae <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0296	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/mass_operator_results.json b97f96fio829 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0297	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/three_part_bubble.py 552e8023a5f7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0298	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/three_part_bubble_ results.json 1co82b7e25eo <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0299	V	CODE	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/verify_review.py 47oe5289b538 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0300	V	CERT	MONOGRAPH/06 HELICAL RELATIVITY/02 COMPUTATIONS/verify_review_results.json fobeb661f792 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0301	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/Bubble_Field_Closure_Attack_G_ Source_Weak_Leptons_Joseph_Shields.docx bb6be69eabfa <i>Historical development and withdrawn field routes; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0302	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/Bubble_Field_Closure_Attack_G_ Source_Weak_Leptons_Joseph_Shields.pdf 3b7dc8a5b6az <i>Historical development and withdrawn field routes; mapped provisional</i>
S0303	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/The_Bubble_Field_Compression_ Boundary_Closure_Joseph_Shields.docx bfd2o6425857 <i>Historical development and withdrawn field routes; mapped provisional</i>
S0304	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/The_Bubble_Field_Compression_ Boundary_Closure_Joseph_Shields.pdf 449f92368667 <i>Historical development and withdrawn field routes; mapped provisional</i>
S0305	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/The_Bubble_Field_Compression_ Boundary_Closure_Joseph_Shields_ REFERENT_PRINCIPLE.docx 86b795f22bbo <i>Historical development and withdrawn field routes; mapped provisional</i>
S0306	V	HIST	MONOGRAPH/06 HELICAL RELATIVITY/90 HISTORY/The_Bubble_Field_Rigorous_ Formulation_Joseph_Shields.docx cfe6o2fi3364 <i>Historical development and withdrawn field routes; mapped provisional</i>
S0307	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/A World of Distinctions.docx 61o7d6azocaf <i>Historical and supersession ledger; mapped provisional</i>
S0308	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/A World of Distinctions.pdf dae4f7b74254 <i>Historical and supersession ledger; mapped provisional</i>
S0309	V	MANIFEST	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ FREEZE MANIFEST.md 36244f5c6b85 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0310	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ PDF Editions/Universopedia – The Observational Standard.pdf aec9f71413f7 <i>Historical and supersession ledger; mapped provisional</i>
S0311	V	NOTE	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ READ ME FIRST.md

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ID	Volume	Role	Source and disposition
S0312	V	LEDGER	o3df56466fao <i>Historical and supersession ledger; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ README.md
S0313	V	LEDGER	efed703445ba <i>Historical and supersession ledger; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/README.md
S0314	V	CERT	c3345f98ce45 <i>Historical and supersession ledger; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/results/interface_ results.json
S0315	V	CERT	9ce7d383b991 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/results/sparc_ results.json
S0316	V	CODE	dcoa39b13ede <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/ Fourth/Research Archive/scripts/dev_ neurogenesis_test.py
S0317	V	CODE	fc7bd86066a9 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/scripts/e8_golden_ tests.py
S0318	V	CODE	6b6af9428foo <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/scripts/interface_ integral_test.py
S0319	V	CODE	d9a342e39f3f <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/scripts/neuro_ avalanche_test.py
S0320	V	CODE	72f8cb5d9a7b <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/scripts/sgz_balanced_ turnover_test.py
			49f3689e9a84 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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Table 9.2 continued from the preceding page

ID	Volume	Role	Source and disposition
S0321	V	CODE	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Research Archive/scripts/sparc_test.py 914aff77bd97 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0322	V	MS	MONOGRAPH/90 ARCHIVE/Printings/ Fourth/Research Archive/Systems of Interpretation – detailed workings/ Application 02 – The Processing-Machine Motif and a Neuronal Test.pdf c21595554458 <i>Historical and supersession ledger; mapped provisional</i>
S0323	V	MS	MONOGRAPH/90 ARCHIVE/Printings/ Fourth/Research Archive/Systems of Interpretation – detailed workings/ Application 04 – The Balanced-Turnover Test (Subgranular Zone).pdf f9c31b4d6c57 <i>Historical and supersession ledger; mapped provisional</i>
S0324	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Unified Mechanics – The Formal Register.docx af4b5bf26509 <i>Historical and supersession ledger; mapped provisional</i>
S0325	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Unified Mechanics – The Formal Register.pdf bf38c95088a6 <i>Historical and supersession ledger; mapped provisional</i>
S0326	V	MS	MONOGRAPH/90 ARCHIVE/Printings/ Fourth/Universopedia – Systems of Interpretation.pdf 9bc3b2494e1c <i>Historical and supersession ledger; mapped provisional</i>
S0327	V	MS	MONOGRAPH/90 ARCHIVE/Printings/Fourth/ Universopedia – The Observational Standard.pdf 58cdc747a166 <i>Historical and supersession ledger; mapped provisional</i>
S0328	V	LEDGER	MONOGRAPH/ADDENDA/00 ADDENDA INDEX.md b68c270752b8 <i>Addenda protocol and provenance; mapped provisional</i>
S0329	V	CODE	MONOGRAPH/ADDENDA/00 hash_addenda.py 61c50edb6fdc <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0330	V	MANIFEST	MONOGRAPH/ADDENDA/A01 The Resolution Operator/SHA256SUMS.txt

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ID	Volume	Role	Source and disposition
S0331	III	MS	475203056bco <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A01 The Resolution Operator/The_Resolution_Operator_Joseph_Shields.docx
S0332	III	MS	f9d9b5b69613 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
			MONOGRAPH/ADDENDA/A01 The Resolution Operator/The_Resolution_Operator_Joseph_Shields.pdf
S0333	V	MANIFEST	4db4bd93cffe <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
			MONOGRAPH/ADDENDA/A02 The Field Derivation Suite/SHA256SUMS.txt
S0334	V	CODE	7d96383f5e59 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A02 The Field Derivation Suite/um_field_suite.py
S0335	V	CERT	ao6d6cb181fe <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A02 The Field Derivation Suite/um_field_suite_results.json
S0336	V	MANIFEST	d770be1bo4b3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A03 The Forward Solver/SHA256SUMS.txt
S0337	V	CODE	43e8d27do812 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A03 The Forward Solver/um_forward_solver.py
S0338	V	CERT	7e944882b29b <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A03 The Forward Solver/um_forward_solver_results.json
S0339	V	CODE	664o7c6fdobc <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/ADDENDA/A04 The Internal State Instrument/internal_state_instrument.py
			1fb3c2db91ba <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0340	V	CERT	MONOGRAPH/ADDENDA/A04 The Internal State Instrument/internal_state_instrument_results.json 5ef4e78f5do3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0341	V	MANIFEST	MONOGRAPH/ADDENDA/A04 The Internal State Instrument/SHA256SUMS.txt 7f24eod516e1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0342	III	MS	MONOGRAPH/ADDENDA/A04 The Internal State Instrument/The_Internal_State_Instrument_Joseph_Shields.docx 17o696dzeb19 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0343	III	MS	MONOGRAPH/ADDENDA/A04 The Internal State Instrument/The_Internal_State_Instrument_Joseph_Shields.pdf 99925f34accc <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0344	V	CODE	MONOGRAPH/ADDENDA/A05 Prediction From The Colour Average/colour_average_prediction.py 722e3a8699ea <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0345	V	CERT	MONOGRAPH/ADDENDA/A05 Prediction From The Colour Average/colour_average_prediction_results.json 9793a4dfa447 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0346	V	MANIFEST	MONOGRAPH/ADDENDA/A05 Prediction From The Colour Average/SHA256SUMS.txt 48fdd62aco68 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0347	V	MANIFEST	MONOGRAPH/ADDENDA/A06 The Whole Thing Once/SHA256SUMS.txt 14a7b14f5eao <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0348	I; II; III; IV; V	MS	MONOGRAPH/ADDENDA/A06 The Whole Thing Once/The_Whole_Thing_Once_Joseph_Shields.docx 35ocbo7bf21c <i>Corpus-wide synthesis and concordance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0349	I; II; III; IV; V	MS	MONOGRAPH/ADDENDA/A06 The Whole Thing Once/The_Whole_Thing_Once_Joseph_ Shields.pdf de8b13f41933 <i>Corpus-wide synthesis and concordance; mapped provisional</i>
S0350	V	CODE	MONOGRAPH/ADDENDA/A07 Helical Multiplication/helical_ multiplication.py 92b803a6b4f3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0351	V	CERT	MONOGRAPH/ADDENDA/A07 Helical Multiplication/helical_multiplication_ results.json 40c4fofa4696 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0352	V	MANIFEST	MONOGRAPH/ADDENDA/A07 Helical Multiplication/SHA256SUMS.txt 8fd212a671d6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0353	III	NOTE	MONOGRAPH/ADDENDA/A08 The Calculator/ NOTES.md d11775a54b47 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0354	V	MANIFEST	MONOGRAPH/ADDENDA/A08 The Calculator/ SHA256SUMS.txt 71a72a366a75 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0355	V	CODE	MONOGRAPH/ADDENDA/A08 The Calculator/ um_assemble.py 6039fe8af54b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0356	V	CERT	MONOGRAPH/ADDENDA/A08 The Calculator/ um_assemble_results.json f70bf799afc4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0357	V	CODE	MONOGRAPH/ADDENDA/A08 The Calculator/ um_calculator.py 4ca8e00034a4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0358	V	CERT	MONOGRAPH/ADDENDA/A08 The Calculator/ um_calculator_results.json e1naccfabd11 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0359	V	CODE	MONOGRAPH/ADDENDA/A08 The Calculator/ um_chain.py 64308636c379 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0360	V	CERT	MONOGRAPH/ADDENDA/A08 The Calculator/ um_chain_results.json 3d700981bbf1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0361	V	CODE	MONOGRAPH/ADDENDA/A08 The Calculator/ um_consensus.py a4a3bb258e47 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0362	V	CERT	MONOGRAPH/ADDENDA/A08 The Calculator/ um_consensus_results.json 012150132f2a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0363	V	CODE	MONOGRAPH/ADDENDA/A08 The Calculator/ um_realisation.py 385ae499842a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0364	V	CERT	MONOGRAPH/ADDENDA/A08 The Calculator/ um_realisation_results.json e2855a9783b4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0365	V	CODE	MONOGRAPH/ADDENDA/A09 The Master Equation/make_a09_plates.py 7a6ad57ceeb6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0366	V	CODE	MONOGRAPH/ADDENDA/A09 The Master Equation/master_equation_cosmology.py 57b1dccadc6c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0367	V	CERT	MONOGRAPH/ADDENDA/A09 The Master Equation/master_equation_results.json 43e908ce677a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0368	III	NOTE	MONOGRAPH/ADDENDA/A09 The Master Equation/NOTES.md 3a4do4631ae3 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0369	V	MANIFEST	MONOGRAPH/ADDENDA/A09 The Master Equation/SHA256SUMS.txt e8b7afob4b25 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0370	III	MS	MONOGRAPH/ADDENDA/A09 The Master Equation/The_Master_Equation_Joseph_Shields.docx f760aeaddcbo <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0371	III	MS	MONOGRAPH/ADDENDA/A09 The Master Equation/The_Master_Equation_Joseph_Shields.pdf db30a4cc5c96 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
S0372	V	NOTE	MONOGRAPH/ADDENDA/ADDENDUM PROTOCOL.md bdf661941799 <i>Addenda protocol and provenance; mapped provisional</i>
S0373	V	CODE	MONOGRAPH/Field Ledger Second Edition/bookbody1.py 8b1f37925d61 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0374	V	CODE	MONOGRAPH/Field Ledger Second Edition/bookbody2.py 1e84a6b9ad9c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0375	V	CODE	MONOGRAPH/Field Ledger Second Edition/bookbody3.py c59aa20052da <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0376	V	CODE	MONOGRAPH/Field Ledger Second Edition/bookbody4.py 69e680f34eb7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0377	V	CODE	MONOGRAPH/Field Ledger Second Edition/bookhead.py 4004bf14e28c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0378	V	CODE	MONOGRAPH/Field Ledger Second Edition/build.py aa14c25d46c6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0379	V	CODE	MONOGRAPH/Field Ledger Second Edition/fl_base.py 102ce0927254 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0380	V	CODE	MONOGRAPH/Field Ledger Second Edition/fl_charts.py

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ID	Volume	Role	Source and disposition
S0381	V	CODE	cf290526d5a3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/Field Ledger Second Edition/fl_diagrams.py
S0382	V	MANIFEST	co4c159fca63 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/Field Ledger Second Edition/SHA256SUMS.txt
S0383	III; IV	MS	o59fo8357c90 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			MONOGRAPH/Field Ledger Second Edition/Unified_Mechanics_The_Field_Ledger_Second_Edition.docx
S0384	III; IV	MS	94bcd5090c5 <i>Master readout, cosmology, and scientific ledger; mapped provisional</i>
			MONOGRAPH/Field Ledger Second Edition/Unified_Mechanics_The_Field_Ledger_Second_Edition.pdf
S0385	V	LEDGER	d74219a60b62 <i>Master readout, cosmology, and scientific ledger; mapped provisional</i>
			MONOGRAPH/README.md
S0386	II	LEDGER	9674b30039a3 <i>Methods and reproducibility; mapped provisional</i>
			SCI/00 START HERE.md
S0387	II	LEDGER	938d5dfoc82d <i>Least-action field; mapped provisional</i>
			SCI/_ADD TO REPO LEAN/06 HELICAL RELATIVITY/00 START HERE/05 LEAN BUILD NOTE.md
S0388	II	LEDGER	e192538cec8c <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			SCI/_ADD TO REPO/06 HELICAL RELATIVITY/00 START HERE/01 README.md
S0389	II	LEDGER	o65eboe4c221 <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			SCI/_ADD TO REPO/06 HELICAL RELATIVITY/00 START HERE/03 OPEN LEDGER ADDITIONS.md
S0390	V	MANIFEST	89e4f8d8a849 <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			SCI/_ADD TO REPO/06 HELICAL RELATIVITY/00 START HERE/04 SHA256SUMS.txt
			b2f8037f8614 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0391	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/audit_closure_attack_ results.json 9af802d6ca1b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0392	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/bubble_field_closure_ attack_results.json 510b382c3857 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0393	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/bubble_field_ results.json ad92fda836b3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0394	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/bubble_field_rigorous_ results.json 15cc582c4ae9 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0395	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/colour_field_closure_ results.json 3edb19764259 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0396	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/exercising_the_field_ results.json 60eac4f31d77 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0397	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/mass_operator_ results.json 828ee83ed150 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0398	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/three_part_bubble_ results.json 8cc825ceef58 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0399	V	CERT	SCI/_ADD TO REPO/06 HELICAL RELATIVITY/ 02 COMPUTATIONS/verify_review_ results.json

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ID	Volume	Role	Source and disposition
S0400	V	LEDGER	249885235360 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/_ADD TO REPO/README.md
S0401	V	CODE	4db82dd9bcoa <i>Methods and reproducibility; mapped provisional</i>
			SCI/ADDENDA/00 hash_addenda.py
S0402	V	CODE	346d4b645de3 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A02 The Field Derivation Suite/um_field_suite.py
S0403	V	CODE	83dfd09e3c47 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A03 The Forward Solver/um_forward_solver.py
S0404	V	CODE	f12fd136cc51 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A04 The Internal State Instrument/internal_state_instrument.py
S0405	III	NOTE	7946b66bf71a <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A08 The Calculator/NOTES.md
S0406	V	CODE	02e272284e88 <i>Resolution, evolution, helical arithmetic, and master readout; mapped provisional</i>
			SCI/ADDENDA/A08 The Calculator/um_assemble.py
S0407	V	CODE	a78df11023e1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A08 The Calculator/um_chain.py
S0408	V	CODE	3ab4a6c522f5 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A08 The Calculator/um_realisation.py
S0409	V	CODE	5f7cf135cd13 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A09 The Master Equation/make_a09_plates.py
S0410	V	CODE	b70268c385dd <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A09 The Master Equation/master_equation_cosmology.py
S0411	II; V	NOTE	2a459aac96cd <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/ADDENDA/A10 The Bubble Field Audit/bubble_audit_output.txt

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ID	Volume	Role	Source and disposition
S0412	V	NOTE	ad4ac3163025 <i>Field audit and reproducibility; mapped provisional</i>
			672eab5cbdo7 <i>SCI/ADDENDA/ADDENDUM PROTOCOL.md</i>
S0413	V	CERT	672eab5cbdo7 <i>Addenda protocol and provenance; mapped provisional</i>
			7e14b76507c2 <i>SCI/Atomic Web/04_Data/atomic_web_elements_118.csv</i>
S0414	V	CERT	7e14b76507c2 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			d3025bb7cdc5 <i>SCI/Atomic Web/04_Data/atomic_web_elements_118.json</i>
S0415	V	CODE	d3025bb7cdc5 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			8ef4defe4334 <i>SCI/Atomic Web/04_Data/render_complete_atomic_web_png.py</i>
S0416	V	CERT	8ef4defe4334 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			809fbe17eccb <i>SCI/Atomic Web/05_Audit/audit_atomic_web_results.json</i>
S0417	V	CERT	809fbe17eccb <i>Reproducibility assets and computational provenance; mapped provisional</i>
			ecc2a535e277 <i>SCI/Atomic Web/05_Audit/audit_results_118.csv</i>
S0418	IV	NOTE	ecc2a535e277 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			9aa442e4b231 <i>SCI/Atomic Web/05_Audit/audit_summary.txt</i>
S0419	V	CODE	9aa442e4b231 <i>Atomic atlas, chemistry provenance, and scope limits; mapped provisional</i>
			f100dbf325a7 <i>SCI/Atomic Web/05_Audit/independent_audit.py</i>
S0420	V	CERT	f100dbf325a7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			ee9bae4c7944 <i>SCI/Atomic Web/manifest.json</i>
S0421	IV	LEDGER	ee9bae4c7944 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			984d48760e2c <i>SCI/Atomic Web/README.txt</i>
S0422	V	MANIFEST	984d48760e2c <i>Atomic atlas, chemistry provenance, and scope limits; mapped provisional</i>
			a99b16391342 <i>SCI/Atomic Web/SHA256SUMS.txt</i>
S0423	III	MS	a99b16391342 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			c1e47568c585 <i>SCI/dl papers/Personal_Relativity_Working_Formalism.docx</i>
			c1e47568c585 <i>Observer resolution and readout; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0424	III	MS	SCI/dl papers/The_Depth_of_Traversal_Joseph_Shields.docx 235e1f4e25ee <i>Traversal and retained history; mapped provisional</i>
S0425	III	MS	SCI/dl papers/Traversal_Depth_Across_AdS_CFT_Joseph_Shields_Revised.docx b110f68a0d2d <i>Traversal and retained history; mapped provisional</i>
S0426	V	CODE	SCI/Field Ledger Second Edition/bookbody1.py 13cc8dcd6da <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0427	V	CODE	SCI/Field Ledger Second Edition/bookbody2.py ob628cde2253 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0428	V	CODE	SCI/Field Ledger Second Edition/bookbody4.py 2c7cd5f55d7b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0429	V	CODE	SCI/Field Ledger Second Edition/build.py fb524da1408c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0430	V	CODE	SCI/Field Ledger Second Edition/fl_base.py 502c7d26f8ee <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0431	V	CODE	SCI/Field Ledger Second Edition/fl_charts.py 57e313c7ab88 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0432	V	CODE	SCI/Field Ledger Second Edition/fl_diagrams.py 5cc1775610f1 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0433	V	CODE	SCI/Helical Relativity/01 Computations/audit_closure_attack.py 9ce880fe6973 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0434	V	CERT	SCI/Helical Relativity/01 Computations/audit_closure_attack_results.json 90c44a9c7b20 <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0435	V	CODE	SCI/Helical Relativity/01 Computations/ bubble_field_closure_attack.py 544b1283100b <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0436	V	CERT	SCI/Helical Relativity/01 Computations/ bubble_field_closure_attack_ results.json 427b3d86532d <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0437	V	CERT	SCI/Helical Relativity/01 Computations/ bubble_field_results.json a128002346ef <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0438	V	CODE	SCI/Helical Relativity/01 Computations/ bubble_field_rigorous.py 38370b3d965e <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0439	V	CERT	SCI/Helical Relativity/01 Computations/ bubble_field_rigorous_results.json c26e648edba6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0440	V	CERT	SCI/Helical Relativity/01 Computations/ colour_field_closure_results.json 796aa5c1da67 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0441	V	CODE	SCI/Helical Relativity/01 Computations/ exercising_the_field.py fa89183aa38f <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0442	V	CERT	SCI/Helical Relativity/01 Computations/ exercising_the_field_results.json 735909432165 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0443	V	CERT	SCI/Helical Relativity/01 Computations/ mass_operator_results.json aa1ff2c44cf4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0444	V	CODE	SCI/Helical Relativity/01 Computations/ three_part_bubble.py 97c7f94c1da7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0445	V	CERT	SCI/Helical Relativity/01 Computations/ three_part_bubble_results.json

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ID	Volume	Role	Source and disposition
S0446	V	CODE	286337d13c23 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/01 Computations/verify_review.py
S0447	V	CERT	9a07f5216455 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/01 Computations/verify_review_results.json
S0448	V	CODE	1765d5dccc0e <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/02 Interactive/universal-field-exoplanet-lab.html
S0449	V	CODE	25a6c0dfc92 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/02 Interactive/universal-field-solar-system.html
S0450	II	NOTE	c352e6deec1e <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/04 Foundations/The Bubble Field(1).md
S0451	III	MS	4ae07297829a <i>Least-action field, geometry, sources, and gravitation; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/00 Paper/The_Observer_Interior_Quantum_Colour_Field_Joseph_Shields.docx
S0452	III	MS	386306865d8e <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/00 Paper/The_Observer_Interior_Quantum_Colour_Field_Joseph_Shields.pdf
S0453	V	CODE	f965505fbc71 <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/01 Field Implementation/observer_interior_quantum_colour_field.py
S0454	V	CERT	192c930a815b <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/01 Field Implementation/observer_interior_quantum_colour_field_results.json
			e7e214c54bdf <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0455	V	CODE	SCI/Helical Relativity/06 Colour Field/ 01 Field Implementation/observer_ interior_quantum_colour_field_ visuals.py 2cce334a90d4 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0456	III	NOTE	SCI/Helical Relativity/06 Colour Field/01 Field Implementation/ requirements.txt a1083azd359b <i>Colour/helical readout fields; mapped provisional</i>
S0457	V	CODE	SCI/Helical Relativity/06 Colour Field/ 01 Field Implementation/run_quantum_ colour_field.command 18o6838fdb50 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0458	V	CODE	SCI/Helical Relativity/06 Colour Field/02 Interactive/observer-interior- quantum-colour-field.html 79d5f668276f <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0459	III	NOTE	SCI/Helical Relativity/06 Colour Field/ 04 Data and Recipes/COLOUR_LANGUAGE_ SPEC.md 513056e53255 <i>Colour/helical readout fields; mapped provisional</i>
S0460	V	CERT	SCI/Helical Relativity/06 Colour Field/ 04 Data and Recipes/object_colour_ recipes.json 117b641a6863 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0461	III	MS	SCI/Helical Relativity/06 Colour Field/05 Dictionary/Quantum_Colour_ Dictionary.docx 16e1da838c2a <i>Colour/helical readout fields; mapped provisional</i>
S0462	V	CERT	SCI/Helical Relativity/06 Colour Field/05 Dictionary/Quantum_Colour_ Dictionary.json ede45c493f7a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0463	III	MS	SCI/Helical Relativity/06 Colour Field/05 Dictionary/Quantum_Colour_ Dictionary.pdf

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ID	Volume	Role	Source and disposition
S0464	V	CERT	20df5219a68d <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/05 Dictionary/Quantum_Colour_Dictionary_Presets.csv
S0465	III	LEDGER	65a380de15cb <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/05 Dictionary/Quantum_Colour_Dictionary_README.txt
S0466	V	CODE	adeccce70df3 <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/colour_field_atlas.py
S0467	V	CERT	366500009491 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/colour_field_atlas_results.json
S0468	III	NOTE	a54113506de8 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/06 Colour Field/HOW TO READ A COLOUR FIELD.md
S0469	III	MS	00c988f7729b <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/07 Helical Field/00_Paper/The_Helical_Bubble_Field_Observer_Interior_Time_Evolution_Joseph_Shields.docx
S0470	III	MS	f278c1aeec5f <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/07 Helical Field/00_Paper/The_Helical_Bubble_Field_Observer_Interior_Time_Evolution_Joseph_Shields.pdf
S0471	V	CODE	98e7d6553e69 <i>Colour/helical readout fields; mapped provisional</i>
			SCI/Helical Relativity/07 Helical Field/01_Field_Implementation/helical_bubble_field.py
S0472	V	CERT	af5e6b264965 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Helical Relativity/07 Helical Field/01_Field_Implementation/helical_bubble_field_results.json
			e1195eacod8b <i>Reproducibility assets and computational provenance; mapped provisional</i>

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ID	Volume	Role	Source and disposition
S0473	V	CODE	SCI/Helical Relativity/07 Helical Field/01_Field_Implementation/helical_bubble_field_visuals.py adc2f3d19a0c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0474	V	CODE	SCI/Helical Relativity/07 Helical Field/01_Field_Implementation/run_helical_field.command 45f48ffab905 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0475	V	CODE	SCI/Helical Relativity/07 Helical Field/02_Interactive_Field/helical-bubble-field.html 7574bdd4efb6 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0476	III	LEDGER	SCI/Helical Relativity/07 Helical Field/README.md 6816a71b225a <i>Colour/helical readout fields; mapped provisional</i>
S0477	V	MANIFEST	SCI/Helical Relativity/07 Helical Field/SHA256SUMS.txt 9f01d748c59c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0478	V	MANIFEST	SCI/Helical Relativity/SHA256SUMS.txt 1437809f401c <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0479	V	CODE	SCI/Informational Least Action/bubble_audit.py 9ceaoe789994 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0480	V	CODE	SCI/Informational Least Action/build.py 89e5ef146f42 <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0481	V	CODE	SCI/Informational Least Action/ila_plates.py 23c75c9bdeai <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0482	V	CODE	SCI/Informational Least Action/ilabody.py 4ec80469c20a <i>Reproducibility assets and computational provenance; mapped provisional</i>
S0483	V	CODE	SCI/Informational Least Action/ilahead.py

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ID	Volume	Role	Source and disposition
S0484	I; II	MS	320ce9ecdae7 <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Informational_Least_Action/Informational_Least_Action_and_the_Least_Boundary_Field.docx
S0485	I; II	MS	44d2090310df <i>Informational action and least-boundary field; mapped provisional</i>
			SCI/Informational_Least_Action/Informational_Least_Action_and_the_Least_Boundary_Field.pdf
S0486	V	MANIFEST	1ef2da47dab3 <i>Informational action and least-boundary field; mapped provisional</i>
			SCI/Informational_Least_Action/SHA256SUMS.txt
S0487	I	MS	e9d6ee9ocf7d <i>Reproducibility assets and computational provenance; mapped provisional</i>
			SCI/Unified_Mechanics_Dual_Rank_Rederivation_2026.docx
S0488	I	MS	1e436456821b <i>Fixed point and three-channel partition; mapped provisional</i>
			SCI/Unified_Mechanics_Dual_Rank_Rederivation_2026.pdf
S0489	I	MS	57dc63f92d36 <i>Fixed point and three-channel partition; mapped provisional</i>
			SCI/Unified_Mechanics_Dual_Rank_Rederivation_2026_with_front_note.pdf
S0490	IV	MS	643866ab2cf6 <i>Fixed point and three-channel partition; mapped provisional</i>
			SCI/Unified_Mechanics_The_Field_Ledger(2).docx
S0491	IV	MS	aa3490bb2538 <i>Cosmology and phenomenology; mapped provisional</i>
			SCI/Unified_Mechanics_The_Field_Ledger.docx
			94d094e82dof <i>Cosmology and phenomenology; mapped provisional</i>

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